TIANHAO CAO

British Columbia, Canada | snalyf.c@gmail.com | snalyf.github.io | LinkedIn | GitHub

OBJECTIVE

M.S. Data Science candidate at UBC (Computational Linguistics) with a strong foundation in machine learning, NLP, LLM/VLM research, RAG systems, and causal inference. Experienced in building end-to-end ML pipelines, retrieval-augmented generation systems, and privacy-focused AI research. Seeking opportunities at the intersection of applied machine learning and language understanding, where rigorous statistical thinking and hands-on engineering drive real-world impact.

EDUCATION

University of British Columbia — British Columbia, Canada
M.S. in Data Science (Computational Linguistics) — *September 2025 – June 2026*
Relevant Coursework: Algorithms & Data Structures, Statistical Inference & Computation, Regression, Feature & Model Selection, Supervised & Unsupervised Machine Learning, Deep Learning, Natural Language Processing, Computational Semantics, Machine Translation, Advanced Corpus Linguistics, Time Series Analysis, Databases & Data Retrieval, Data Visualization

University of California, Santa Cruz — California, USA
B.A. in Business Management Economics — *September 2019 – June 2022*
Relevant Coursework: Advanced Quantitative Methods, Time Series, Machine Learning, Statistics, Econometrics, Corporate Finance, Accounting, Marketing, Security Market, Managerial Cost Accounting

TECHNICAL SKILLS

Programming	Python, R, SQL, MongoDB, Bash
ML & AI	PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, LLM Fine-tuning, RAG Pipelines, Vector Embeddings, Prompt Engineering, Multi-agent LLM, SHAP, VLM/CLIP
NLP	spaCy, SentenceTransformers, BERT/RoBERTa, Flan-T5, BiLSTM, jieba, Whoosh, ChromaDB, BM25s, LangChain, Ragas
Web & API	FastAPI, REST API Design, Pydantic, Gradio
Data Engineering	ETL Pipelines, Feature Engineering, Pandas, NumPy, Apache Parquet, Data Visualization
MLOps & Cloud	Docker, AWS, Git/GitHub, CI/CD, Weights & Biases (W&B)
3D & Creative	Blender, FL Studio
Languages	English (Professional Working), Mandarin (Native)
Work Auth.	Canadian Post-Graduation Work Permit (PGWP) Eligible

EXPERIENCE

Applied Machine Learning Engineer Intern — British Columbia, Canada
Ednoda × UBC MDS-CL Capstone — *2026*

- Built an end-to-end AI-powered curriculum content pipeline for an ESL edtech platform, spanning grammar analysis, semantic retrieval, and automated teaching-material generation.
- Designed 29 spaCy `DependencyMatcher` grammar templates to classify ESL sentences by syntactic structure, achieving **65.8% coverage** on production data and 54.2% on the CEFR-SP benchmark; fine-tuned a BERT model for CEFR difficulty classification (A1–C2) reaching **92% (±1) accuracy**.
- Built an offline embedding pipeline (`all-MiniLM-L6-v2`) over 6,477 grammar-tagged sentences and a three-layer retrieval system fusing grammar filtering, CEFR matching, and cosine-similarity ranking to surface curriculum-aligned examples.
- Engineered a three-layer content generator (RoBERTa for single-token slots, T5/Flan-T5 for multi-token spans) and a multi-agent LLM workflow that ingests a textbook PDF and automatically produces a complete teaching package — lesson notebook, practice questions, and graded quiz.

PROJECTS

Membership Inference Attack on Fine-tuned Language Models UBC
LLM Privacy Research — Kaggle Competition *GitHub*

- Achieved **AUC 0.907 and TPR@FPR=0.1 of 0.697** on the validation split — a **+55% AUC** and **+349% TPR** improvement over the raw-loss baseline — the top-performing method in the competition.
- Designed an Improved Neighborhood Attack using dual-model feature fusion across four models (fine-tuned SmolLM2 135M/360M plus base checkpoints), capturing memorization signals invisible to single-model baselines.
- Engineered a 16-dimensional feature vector spanning fine-tuned losses, base losses, reference ratios, neighborhood perturbation gaps, and cross-model agreement, feeding a Gradient Boosting classifier (300 estimators); replaced slow BERT mask-and-fill with fast token-drop perturbation while maintaining quality.

Membership Inference Attacks on Vision-Language Models Independent
Multimodal Privacy Research *GitHub*

- Achieved best result of **AUC 0.849 and TPR@FPR=0.1 of 0.388** by fusing loss-ratio and neighborhood-perturbation features via Logistic Regression, progressing from a raw-loss baseline (AUC 0.50) across five systematically designed attack variants.
- Adapted the M^4I framework (Hu et al., NeurIPS 2022) to a CLIP + SmolVLM-256M architecture, replacing legacy ResNet-152/LSTM shadow models with CLIP ViT-B/32 as an off-the-shelf cross-modal feature extractor — **Kaggle AUC 0.8056** with no shadow-model training.
- Extended MIA to the multimodal neighborhood setting (text dropout + image Gaussian noise, patch masking, random crop), extracting 17 cross-modal sensitivity features; found CLIP image-text cosine similarity dominates the membership signal (48% of feature importance).

MDS-CL RAG Assistant Independent
Domain-Specific RAG Chatbot *GitHub*

- Built an end-to-end RAG chatbot grounded in the UBC MDS-CL program website: scraped and chunked HTML (Markdown-header + 800-char recursive splitting) into a dense ChromaDB store (`all-MiniLM-L6-v2`) and a BM25s sparse index, fused via Reciprocal Rank Fusion (k=60).
- Integrated Gemini 2.5 Flash via an OpenAI-compatible endpoint with a strict grounding prompt; added an intent-classification layer with per-intent few-shot injection that short-circuits out-of-scope queries before retrieval.
- Evaluated the pipeline with Ragas (Faithfulness, AnswerRelevancy, ContextRecall, ContextPrecision) on auto-generated Q&A pairs, and shipped a Gradio chat UI with live-reconfigurable API settings and streaming output.

Full-Stack Corpus Search Engine with LLM Annotation UBC
Bilingual Search + ML Annotation (4-person team) *GitHub*

- Co-built a bilingual (English + Mandarin) full-text search engine with FastAPI and Whoosh; implemented a custom jieba-based `ChineseAnalyzer` for accurate segmentation, enabling sub-second multi-language queries, exposed via 7 Pydantic-validated REST endpoints.
- Owned a 3-stage annotation pipeline: bilingual TF-IDF + LinearSVC pseudo-labeling, Active Learning prioritization (Query-by-Uncertainty + Query-by-Committee), and few-shot data augmentation to expand the labeled set.
- Trained and compared neural transfer-learning models under different freezing strategies plus a multi-task variant jointly optimizing fraud classification and silver NER; containerized the full application with Docker for reproducible builds.

Betty Talker — Recipe RAG Pipeline UBC
Academic Project *GitHub*

- Built a full RAG pipeline from scratch: parsed a public-domain recipe book (Project Gutenberg) into structured JSON via regex extraction, then generated `all-MiniLM-L6-v2` embeddings to construct a searchable vector store with cosine-similarity retrieval.

- Designed a few-shot LLM intent classifier routing queries into recipe search (RAG), ingredient scaling, or off-topic refusal — an action-aware agent grounded strictly in retrieved context.
- Built a recipe-scaling module supporting Unicode fractions; packaged the pipeline as a modular Python project with a CLI interface, `.env`-based key management, and a one-command setup script.

Yelp Review Sentiment Analysis Independent
NLP / Deep Learning Pipeline *GitHub*

- Built a fully modular ML pipeline (`data_loader`, `eda`, `preprocessing`, `models`, `train`) for 5-class star-rating classification, replacing a monolithic notebook with clean, reusable components.
- Compared three progressively complex models — TF-IDF + Logistic Regression baseline, a CBOW classifier over pretrained spaCy vectors, and a 3-layer Bidirectional LSTM with `pack_padded_sequence` — reaching best validation **Macro F1 of 0.557** on 27,500 reviews.
- Implemented a generic training loop with F1-based early stopping (patience=3), checkpointing, and per-epoch loss/F1 reporting.

Credit Card Default Analysis UBC
End-to-End ML Pipeline *GitHub*

- Trained and evaluated multiple classifiers (Logistic Regression, Random Forest, XGBoost) on a 30,000-sample UCI dataset, selecting the optimized XGBoost model for the best Precision/Recall balance (**F1-score 0.55**) on a heavily imbalanced dataset.
- Conducted extensive EDA on data distributions and class imbalance; applied resampling and feature scaling to improve model robustness across the full pipeline.
- Used SHAP values to interpret model decisions, identifying recent payment history as the most critical default predictor and translating outputs into actionable business insights.

Predictive Modeling of Agricultural Insurance Purchase Behavior UC Santa Cruz
Causal / Econometric Modeling *GitHub*

- Achieved **76% predictive precision** by developing and comparing Logit (baseline), Lasso (feature selection), and Random Forest models on a cleaned, merged survey dataset of 1,000+ records and 30+ variables from the Harvard Dataverse.
- Interpreted model outputs via feature-importance metrics to identify key behavioral drivers and translate technical findings into business insights.
- Conducted error analysis to assess prediction variance, discussing model limitations and potential survey biases.

Difference-in-Differences Analysis of COVID-19 Policy Impacts UC Santa Cruz
Causal Inference / Panel Data *GitHub*

- Implemented a Difference-in-Differences framework in R to quantify the causal impact of mandatory vs. optional lockdown policies on COVID-19 infection and mortality rates across 5 major economies (US, China, Japan, UK, India).
- Aggregated and standardized cross-country panel data from the WHO and Our World in Data; mitigated omitted-variable bias with control variables to improve robustness and validity of the causal estimates.
- Performed robustness checks against baseline models to validate policy implications and discuss potential confounding factors.